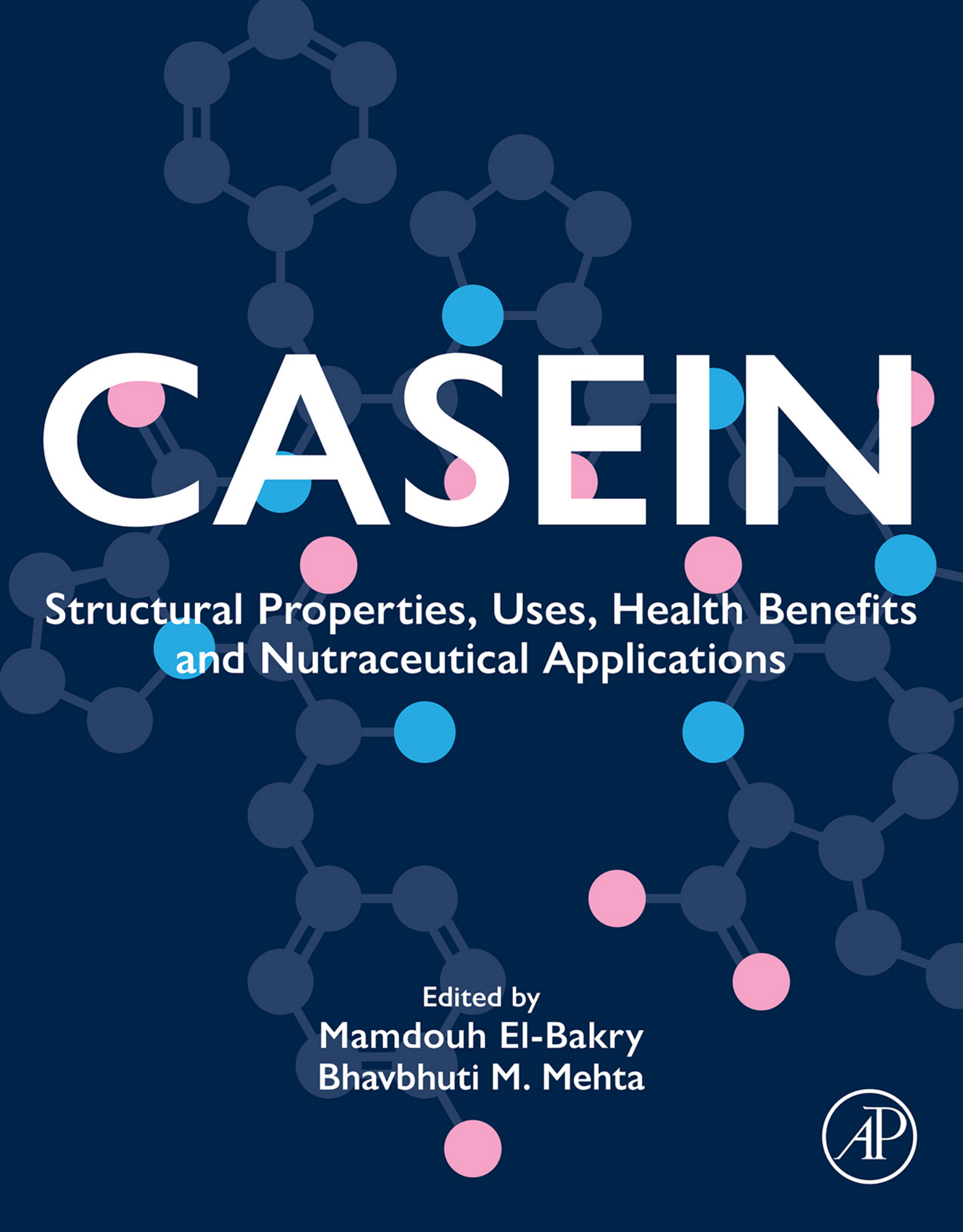


The background of the cover features a complex molecular structure graphic. It consists of numerous dark blue spheres connected by thin lines, forming a network of hexagonal and other geometric shapes. Interspersed within this network are several bright blue and pink spheres, which appear to be part of the molecular structure or represent specific atoms or functional groups. The overall effect is a scientific and modern aesthetic.

CASEIN

Structural Properties, Uses, Health Benefits
and Nutraceutical Applications

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Casein polymorphisms and implications in cheesemaking

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12.1 Introduction

Cheese is a fermented food produced from milk of mammal animals such as sheep, buffalo, cows, and goats (Beresford, 2023). Its production process marked an important step during the Neolithic (7th millennium BC) for allowing to concentrate the high-valuable nutrients of milk in a product suitable for young and old people and easy to preserve in comparison to raw milk (Kindstedt, 2012). Although almost all cheeses include these four ingredients: milk, microbes, proteases, and salt, there is a vast variety of them around the world (> 1000) with different sensory attributes and nutritional value, such as Brie, Cheddar, Danish Blue, Mozzarella, and Parmesan (Beresford, 2023; Beresford et al., 2001). This variation is a consequence of differences in formulation (chemical substances present in cheese), composition, and production process (McSweeney et al., 2017). Formulation and composition can be modified by substances added to cheese during the manufacturing process and by the components contained in the origin milk. These components, that include mainly water, proteins (mainly caseins [CNs] and whey proteins), lactose, fat, and minerals (Kelly, 2007), are highly dependent on the physiologic characteristics of the animal (i.e., breed, age, season, etc.) (Bittante et al., 2021).

From all the components of milk, CNs are the most influential proteins on the final physical attributes of cheese, involving variables such as the amount of CN, degree of proteolysis during preparation, polymorphisms, and CN posttranslational modifications (PTM)

(Kamath et al., 2022; Gislon et al., 2023; Holland, 2009). Throughout this chapter, CN polymorphisms, cheese preparation process, and the effects of polymorphisms on cheesemaking are explained. The understanding of CN polymorphisms and their relationship with cheesemaking are of key importance in order to keep and improve the quality of each cheese variety, and also helps improve the yield and economic balance of their respective production processes (Olsen et al., 2023).

12.2 Description of casein polymorphisms

A DNA polymorphism is the coexistence of at least two different alleles at the same locus with a frequency greater than 1% (Forrester et al., 2016). This proportion is intended to differentiate polymorphisms from mutations, as mutations refer to rare occurrences of the DNA sequence, whereas polymorphisms are DNA variations linked to the features or conditions of a population (Karki et al., 2015). Due to the prevalence of particular polymorphisms in distinct populations can be used as genetic markers and to identify DNA variants that may influence the phenotypic expression of several genes (Forrester et al., 2016; Farrell et al., 2004a). Gene polymorphisms involve variations in nucleotide sequences that might result in coding differences in synthesized proteins, such as indels (insertion-deletions) and amino acid substitutions (Calladine et al., 2004; Lisson et al., 2013). Eventually, these amino acid sequence modifications could affect the proteins' biological functioning (Calladine et al., 2004). The different versions of a

protein whose changes were caused by gene polymorphisms are called “protein variants” (Lisson et al., 2013). Some protein variations in the human body are associated with diseases such as influenza, diabetes, epilepsy, Parkinson’s, and various types of cancer (Zhang et al., 2021; Bala et al., 2021; She et al., 2022; Wang et al., 2020; Su et al., 2015).

The polymorphic genetic variations in animals induce alterations in the metabolism that may influence the quality of different products used in the food industry, affecting key factors such as productivity and resistance to diseases (Soloshenko et al., 2016). Some protein variations, for instance, may affect an animal’s development rate, weight, intramuscular fat content, water-retention capacity, sensory qualities, and meat texture (Grochowska et al., 2021) (Greguła-Kania et al., 2019). Protein variations in milk have been linked to yield, fat and protein content, composition, sugar and calcium levels, density, and flavor (Soloshenko et al., 2016). Polymorphisms directly influence all these properties in the genes responsible for encoding CNs, the most abundant group proteins of milk.

The biological function of CNs is to transport calcium phosphate through milk from the mother to the neonate, and each of the CN fractions (i.e., α 1-CN, α 2-CN, β -CN, and κ -CN) has a role in this process (Nadugala et al., 2022). Variations in protein fractions may result from PTMs such as phosphorylation and glycosylation, as well as polymorphisms (Nadugala

et al., 2022). Moreover, the molecular structure of each CN protein may have slight variations in the molecular structure depending on the species (Li et al., 2022). For instance, the bovine mature β -CN is two amino acids shorter than its mature caprine counterpart (Li et al., 2022). Due to these amino acid sequence differences in CNs for different species, the high number of mammal animal species, and the fact that each sequence variation in proteins influences to some extent, their biological functions, the studies on CN polymorphisms must be framed in the species of interest. Bovines have been utilized more than any other mammalian species to produce milk for human consumption. The genetic variability of bovine CN polymorphisms has been investigated for over 60 years, resulting in a significant body of literature devoted to their comprehension (Minetto et al., 2022; Caroli et al., 2009). For these reasons, bovine CN polymorphisms are described in this work to elucidate their properties and, in turn, how they impact the performance of industrial products.

The CN protein fractions α 1-CN, α 2-CN, β -CN, and κ -CN, are encoded by the genes *CSN1S1*, *CSN1S2*, *CSN2*, and *CSN3* of bovine species, respectively (Fig. 12.1) (Nadugala et al., 2022). Consequently, each fraction has its own polymorphisms depending on the nucleotide modifications of each gene. The CN variants include 10 for α 1-CN (A, B, C, D, E, F, G, H, I, and J), 4 for α 2-CN (A, B, C, and D), 12 for β -CN (A^1 , A^2 , A^3 , B, C, D, E, F, G, H^1 , H^2 , and I), and 12 for κ -CN (A, B, B^2 , C, E, F^1 , F^2 , G^1 , G^2 , H, I, and J) (Lisson et al.,

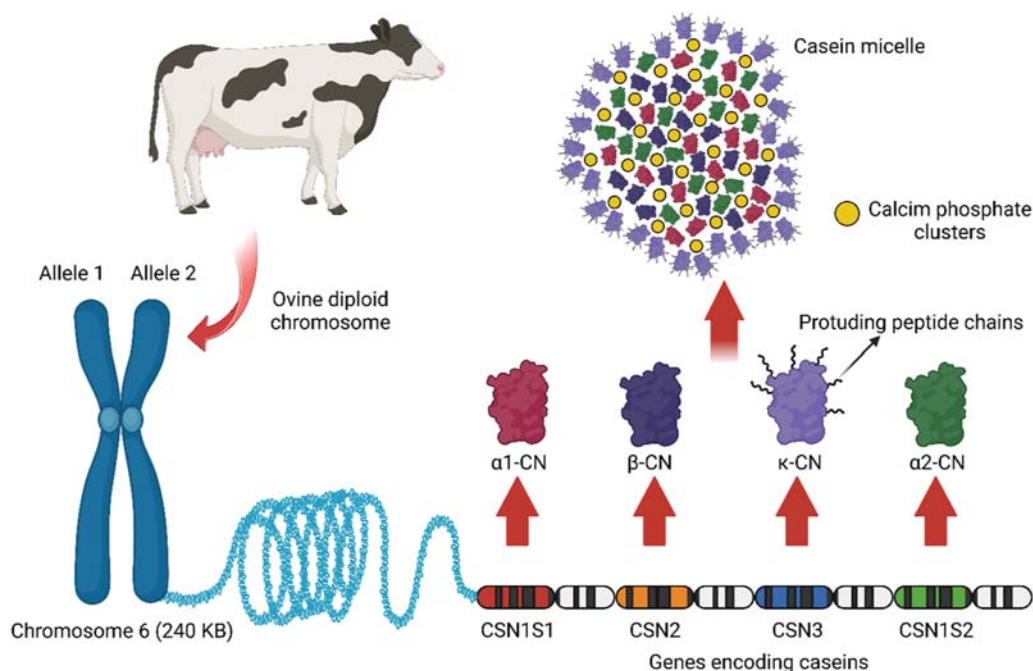


FIGURE 12.1 Schematic representation of an ovine chromosome, parts of the genes that encode each CN fraction, and the structure of a CN micelle in terms of the distribution of each fraction through the micelle.

TABLE 12.1 Sequence of each bovine CN fraction (i.e., α 1-CN, α 2-CN, β -CN, and κ -CN). The sequence shown corresponds to one of the most commonly found variants.

CN fraction	Amino acid sequence
α 1-CN B	RPKHPKHQGLPQEVLENLRRFFVAPFPEVFGKEKVNELSKDIGSESTEDQAMEDIKQMEAESISSSEEIVPNSVEQKHIQ KEDVPSERYLGYLEQLLRLLKKYKVPQLEIVPNSAEERLHSMKEGIHAQQKEPMIGVNQELAYFYPELFRQFYQLDAYPSG AWYYVPLGTQYTDAPSFSDIPNPIGSENSEKTTMPLW
α 2-CN A	KNTMEHVSSEESIISQETYKQEKMAINPSKENLCSTFCKEVVRNANEEFYSIGSSSESAEVATEEVKITVDDKHYQKAL NEINQFYQKFPQYLQYLYQGPIVLNPWDQVKRNAVPITPTLNREQLSTSEENSKKTVDMESTEVFTKTKLTEEEKNRLNF LKKISQRYQKFALPQYLKTVYQHQAAMKPWIQPKTKVIPYVRYL
β -CN A ²	RELEELNVPGEIVESLSSEESITRINKKIEKFQSEEQQQTEDELQDKIHFAQTQSLVYFPFGPIPNLQNIPLTQTPTVVVPP FLQPEVMGVSKVKEAMAPKHKEMPPFKYPVEPFTEQSLTLTDVENLHLPLLLQSWMHQPHQPLPPTVMFPPQSVLSLS QSKVLVPVQKAVPYPQRDMPIQAFLLYQEPVLGPVRGPFPIIV
κ -CN A	QEQNQEPIRCEKDERFFSDKIAKYIPIQYVLSRYSYGLNYYQQKPVALINNQFLPYPYAKPAAVRSPAQILQWQVLSNTVP AKSCQAQPTTMARHPHPLSFMaipPKKNQDKTEIPTINTIASGEPTSTPTTEAVESTVATLEDSPEVIESPEINTVQVTSTAV

2013; Nadugala et al., 2022; Bijl et al., 2020). The variant names are mainly assigned by order of discovery of each polymorphism and employ alphabet letters starting with A (Table 12.1).

12.2.1 α 1-Casein polymorphisms

As depicted in Fig. 12.1, a proportion of α 1-CN is present in all CN micelle volumes and accounts for approximately 40% of them (Dalglish and Corredig, 2012). The first five α 1-CN variants (i.e., A, B, C, D, and E) are the most prevalent; the last five (i.e., F, G, H, I, and J) have been only detected in some cattle breeds (Nadugala et al., 2022). The nomenclature of α 1-CN variants was initially determined based on the relative mobility difference in a starch gel electrophoresis (SGE) assay (Thompson et al., 1965). In this regard, variants A, B, and C were named in this order for their relative mobility values of 1.18, 1.10, and 1.07, respectively (Thompson et al., 1965). Although the variant D had intermediate mobility between A and B (1.14), this letter was assigned in chronological order by Grosclaude et al. (1966) in 1966, 4 years after the discovery of α 1-CN variants A, B, and C, in 1962 (Vaugh et al., 1962) and 2 years after naming them in 1964 (Thompson and Kiddy, 1964). The following variants (i.e., F, G, H, and I) were also termed in chronological order. The variant F was identified in 1993 by isoelectric focusing on polyacrylamide gels and polyacrylamide gel electrophoresis (Erhardt, 1993), while the variant G from 1992 to 1994 by endonucleases (Rando et al., 1992, 2022). Additionally, variant H was identified in 1999 by isoelectric focusing (Mahé et al., 1999) and variant I in 2009 by isoelectric focusing and polymerase chain reaction (PCR) (Lühken et al., 2009). The variant α 1-CN A has been identified in Holstein Friesians, German Red, and Red Holsteins

cattle (Farrell et al., 2004a). This variant is produced by skipping exon IV caused by a mutation in the position +6 of the exon (Mohr et al., 1994). Variant α 1-CN B is the reference and the most common and predominant in *Bos taurus*. Variant C is common in *Bos grunniens* and *Bos indicus*, D in French and Italian breeds and Jerseys in The Netherlands, and E in *B. grunniens* (Farrell et al., 2004a). Variants F and G were found in German Black and White cattle and Italian Brown cows, respectively (Farrell et al., 2004a). As shown in Table 12.2, the variant α 1-CN G has an identical amino acid sequence to B; however, the gene expression is different for both proteins (Nadugala et al., 2022). Finally, the variant α 1-CN J has not been sequenced (Nadugala et al., 2022). One of the most critical impacts of the amino acid changes on α 1-CN variants is the modification of the phosphorylation groups. This is the case of variants D and F, in which one group is added (i.e., threonine) and one group is removed (i.e., serine), respectively (Bijl et al., 2020). Although some studies show that these changes may produce alterations in the internal structure and, therefore, the stability of the CN micelles (Bijl et al., 2014a; Cohen, 2002), the direct relation between phosphorylation degrees and the properties of milk for technological application still requires considerable research (Fang et al., 2018). Regarding the allergenicity, the digestion of the variants α 1-CN B and α 1-CN C showed the production of different peptides related to the substitution 192(E:G) of variant C (Lisson et al., 2014). Some of these peptides were identified as fractions of immunoglobulin E (IgE)–binding epitopes that may be related to allergenicity (Chatchatee et al., 2001). As for the digestion analysis conducted for variants B and C, studying other variants may provide valuable insights into the effects of amino acid changes on allergenicity.

TABLE 12.2 Protein variants for each of the CN fractions.

Protein	Variant	Variation	
		Deletion	Substitution
α 1-CN	A	14–26	
	B	Reference variant	
	C		192(E:G)
	D		53(A:T)
	E		59(Q:K), 192(E:G)
	F		66(S:L)
	G	Identical to the reference variant	
	H	51–58	
	I		84(E:D)
	J	Not available	
α 2-CN	A	Reference variant	
	B		8(S:F)
	C		33(E:G), 47(A:T), 130(T:I)
	D	51–59	
β -CN	A ¹		67(P:H)
	A ²	Reference variant	
	A ³		106(H:Q)
	B		67(P:H), 122(S:R)
	C		37(E:K), 67(P:H)
	D		18(S:K)
	E		36(E:K)
	F		67(P:H), 152(P:L)
	G		67(P:H), 138(P:L)
	H ¹		25(R:C), 88(L:I)
	H ²		72(Q:E), 93(M:L), 114–169(Q:E)
	I		93(M:L)
κ -CN	A	Reference variant	
	B		136(T:I), 148(D:A)
	B ²		136(T:I), 148(D:A), 153(I:T)
	C		97(R:H), 136(T:I), 148(D:A)
	E		155(S:G)
	F ¹		148(D:V)
	F ²		10(R:H)
	G ¹		97(R:C), 135(T:I)
	G ²		148(D:A)
	H		135(T:I)
	I		104(S:A)
	J		136(T:I), 148(D:A), 155(S:R)

12.2.2 α 2-Casein polymorphisms

The fraction α 2-CN represents about 10% of the total CN content and, similarly to α 1-CN, is present in all the volume of the CN micelles (Fig. 12.1) (Hammam et al., 2021). Another function of α 2-CN is to link with the phosphate micro micelles because α 2-CN is the fraction with the highest number of phosphorylation groups (Hammam et al., 2021). The CN micelle stabilization mechanism of α 2-CN also involves the formation of disulfide bonds between cysteine residues located within the same protein (i.e., intramolecular) and residues present in neighboring proteins (i.e., intermolecular) (Bijl et al., 2020; Thorn et al., 2021). The variants of the α 2-CN can be obtained and named according to their mobilities in a gel electrophoresis assay (Eigel et al., 1984). The variant D is the only α 2-CN variant that shows amino acid deletions. This may be attributed to the exon VIII skipping caused by the transversion G $\rightarrow$ T in the last nucleotide of this exon (Bouniol et al., 1993). An additional variant to the four shown in Table 12.2 was previously reported, namely the α 2-CN E (Gallinat et al., 2013); however, its study is not comprehensive enough, and actual references skip it from the α 2-CN variants (Bijl et al., 2020; Nadugala et al., 2022; Gai et al., 2021). Changes in the phosphorylation sites have also been identified in α 2-CN variants. Sixteen phosphorylation sites are present in the reference protein (i.e., α 2-CN A), and they are modified by the substitutions and deletions produced in variants B, C, and D. Results indicate that some of the phosphorylation sites may not be bonded to phosphates. However, the exact identification of each vacant site still requires further study (Bijl et al., 2020).

12.2.3 β -Casein polymorphisms

The fraction β -CN constitutes 33%–45% of the CN bovine milk (Li et al., 2022). Some domains of β -CN can bind to calcium phosphate within the CN micelles, preventing precipitation by supersaturation of the solution (Little and Holt, 2004). Other domains of β -CN are exposed to the internal medium and have been related to the stabilization of water inside the CN micelles (Dalgleish and Corredig, 2012). β -CN may present 12 variants with different sequences (Table 12.2). The β -CN variants A¹, A², and B, together, represent 96% of the allele frequency in Italian Holstein Fresian (Sebastiani et al., 2020). In general, variants A² and A¹ are the most common variants found in bovine animals (Farrell et al., 2004b). A² is considered the original variant, and A¹ comes from mutations in ancient European cattle (Brooke-Taylor et al., 2017). A¹ is only present in bovine milk but

absent in Asian and African breeds (Caroli et al., 2016; Brooke-Taylor et al., 2017).

Even though the only difference between the sequences of A² and A¹ variants is the presence of a histidine instead of proline in amino acid 67 for variant A¹, this change induces numerous health outcomes that have garnered more attention than any other β -CN variants due to their high frequency. Milk with allele A¹ has been associated with type-1 diabetes, coronary heart disease, autism, arteriosclerosis, and schizophrenia (Sodhi et al., 2012). However, the correlation between these diseases and the alleles A¹ and A² still requires more robust experimental evidence (Daniloski et al., 2021). The effect of milk with the alleles A¹ and A² on human gastrointestinal disorders has been shown by different studies in which variant A¹ was related to inflammatory gastric effects (Ho et al., 2014; Jianqin et al., 2016). The reason is that the histidine in the amino acid 67 of variant A¹ is susceptible to proteolytic cleavage; therefore, its digestion generates opioid peptides such as BMC-7, a sequence related to physiological inflammatory responses (Brooke-Taylor et al., 2017). The in vitro cleavage of the variants A¹, A², and B, by pepsin, chymotrypsin, and trypsin, have shown to produce IgE-binding epitopes that contain the sequence variations typical of each allele (Table 12.2) (Lisson et al., 2013). These variations may eventually modify the immunoreactivity of the epitopes and then produce milk with a different effect on gastrointestinal characteristics.

Concerning the terminology, three variants were initially discovered by paper electrophoresis in basic conditions (pH 7.15) using urea, and the order of letters was assigned by the mobility of the variants (i.e., A > B > C) in 1961 (Aschaffenburg, 1961). When electrophoresis was developed in acidic conditions, the variants A¹, A², and A³, were identified and named in order of mobility in the polyacrylamide gel in 1966 (Gai et al., 2021; Peterson and Kopfler, 1966). The variant D was later identified in Zebu cattle with slower mobility than the variant C by gel electrophoresis in alkaline conditions (Aschaffenburg et al., 1968). The variant E was identified by SGE in 1971 (Voglino, 1972), and F and G by reversed-phase high-performance liquid chromatography (HPLC) in 1995 (Visser et al., 1995) and 1998 (Dong and Ng-Kwai-Hang, 1998), respectively. The letter H was simultaneously used to name two different polymorphisms; therefore, they were called H¹ and H² (Farrell et al., 2004a). Variant H¹ was separated by acidic SGE (Han et al., 2000) in 2000, while variant H² was identified by liquid chromatography–mass spectroscopy (LC–Ms) in 2002 (Senocq et al., 2002). The variant β -CN H² has two identified substitutions, as shown in Table 12.2. However, an additional substitution of an amino acid Q for E exists between amino

acids 114–169, whose exact location is unknown (Farrell et al., 2004a). Variant I was identified in different European cattle breeds (i.e., Italian Holstein Friesian, Italian Red Pied, Piemontese, and Angler) in 2002 by polymerase chain reaction single-strand conformation polymorphism (PCR-SSCP) (Jann et al., 2002). Additional β -CN variants A⁴, Bz, J, K, and L, have also been identified. However, their amino acid sequences have not been completely defined in the literature (Nadugala et al., 2022).

12.2.4 κ -Casein polymorphisms

The fraction κ -CN represents about 12% of the total CN content of bovine milk (Fox and Brodtkorb, 2008). Although many models have been proposed to understand the structure of the CN micelles, almost all the literature agrees with the location of κ -CN in the outer surface of the CN micelles, as shown in Fig. 12.1 (Bhat et al., 2016). The κ -CN stabilizes CN micelles through an effect similar to the oil droplet stabilization produced by surfactants in emulsions, and their abundance in milk has been associated with the CN micelle diameter (Fox and Brodtkorb, 2008; Bhat et al., 2016). Concerning the κ -CN variants, A, B, and H are the most common, while C, E, and G² are the less common, and the rest are not often found and are only present in some specific breeds (Lisson et al., 2013). Milk with alleles AA has shown CN micelles with a larger diameter (201.2 ± 10.3 nm) than milk with BB (173.0 ± 9.7 nm) (Bijl et al., 2014b). The same experiments show that a lower total content of glycosylated κ -CN can be correlated to a higher CN micelle average diameter. Thus, it is not clear if micelle diameter differences are specifically related to the presence of variants A and B or a higher content of total glycosylated κ -CN.

Variants A, B, and E have been shown to release IgE-binding epitopes after duodenal and gastric digestion (Lisson et al., 2013). These peptides contain the amino acid substitutions associated with variants A, B, and E (Table 12.2); therefore, an alteration in the allergenic potential of milk containing these variants is expected to be described in future studies. In silico digestion of κ -CN variants B, C, F¹, and G², with the enzymes pepsin, trypsin, chymotrypsin A, chymotrypsin C, pancreatic elastase I, pancreatic elastase II, oligopeptidase B, and oligopeptidase II has produced peptides with lowering blood pressure capabilities (Weimann et al., 2009). However, a recent study has shown no generation of angiotensin I converting enzyme (ACE) inhibitory peptides (Petrat-Melin et al., 2016) by applying an in vitro digestion of κ -CN variants A, B, and E. Additionally, variant A showed a

higher antioxidant capacity than variants B and E (Petrat-Melin et al., 2016). Concerning the nomenclature of κ -CN variants, variants κ -CN A and B were distinguished through a SGE assay at pH 8.6 in 1964, where variant A showed greater mobility than variant B (Neelin, 1964; Mackinlay et al., 1966). Variants C, D, and E were identified by isoelectric focus in 1979, 1987, and 1989, respectively (Di Stasio and Merlin, 1979; SEIBERT et al., 1987; Erhardt, 1989). In 1989, it was discovered that variants C and D were identical (Farrell et al., 2004a); therefore, variants C and D were only called C. Hence, variant D is not present in Table 12.2 (Farrell et al., 2004a). Variants F¹ and F² were named chronologically after variant E. Initially, two authors working independently discovered variant F in 1992 and 1996, using PCR and isoelectric focusing on polyacrylamide gels, respectively (Sulimova et al., 1992; Ikonen et al., 1996). Despite this time difference, the authors in 1996 seemed not to have acknowledged the assignation of the letter F to the new variant discovered 4 years before (Ikonen et al., 1996). Thus, when the characterization of variant F (1996) was developed, and the difference in the molecular structure to variant F (1992) was observed using PCR (Prinzenberg et al., 1996), it was decided to assign superscripts 1 and 2 (i.e., F¹ and F²) in chronological order to differentiate both variants without changing the original letter used by the discoverers (Farrell et al., 2004a). Like variants F¹ and F², variants G¹ and G² were initially named G by different authors in 1996. Erhardt (1996) used isoelectric focusing in polyacrylamide gels and alkaline polyacrylamide gel electrophoresis, and Sulimova et al. (1996) using PCR. To differentiate both variants keeping the same letter, Erhardt's was named G¹ and Sulimova's G² (Farrell et al., 2004a). Variants H and I were discovered in 1999 using DNA sequencing and named in chronological order with respect to G variants (Prinzenberg et al., 2009). Variant J was found later in the same year by RP-HPLC (Mahé et al., 1999). Finally, variant B² was reported in 1987, and its name was assigned, considering its high similarity to variant B (Gorodetskiĭ and Kaledin, 2022).

12.3 Cheese production process

Before detailing the effect of CN polymorphisms on cheesemaking, it is crucial to describe each process step, as depicted in Fig. 12.2.

12.3.1 Standardization of milk

The cheese production process and properties are influenced by the origin of milk constituents,

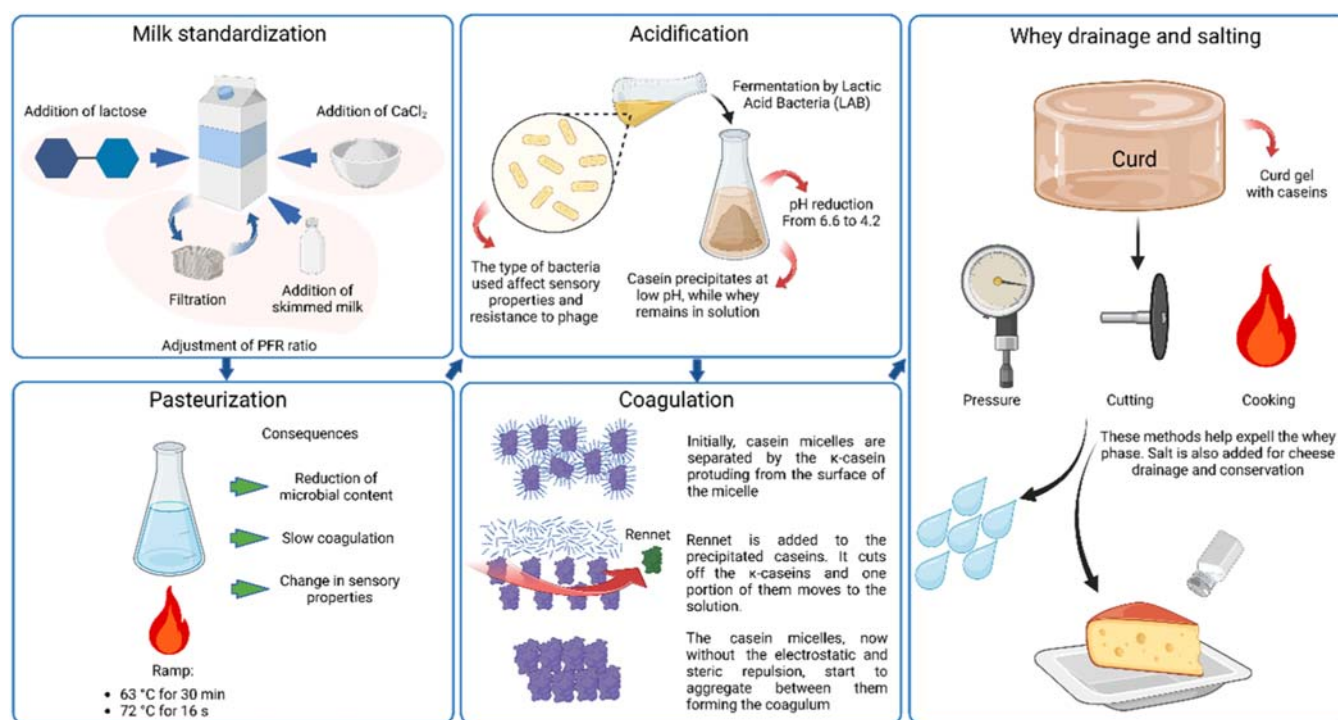


FIGURE 12.2 Schematic representation of the cheesemaking process, variables, and explanation of phenomena associated.

compositions, and relative abundance of their fractions (McSweeney, 2007). The milk substances most related to cheesemaking are fat, CN proteins, lactose, and calcium (McSweeney, 2007). For instance, a high-fat content affects cheese attributes since it helps retain whey and water during curd treatment, resulting in a weaker cheese that ripens more quickly and efficiently (Kalit et al., 2021; Rako et al., 2019). In contrast, low-fat contents are related to doughy cheeses with high firmness and elasticity (Kalit et al., 2021). The protein-to-fat ratio (PFR) produces changes in cheese composition, yield, flavor, rheology, and organoleptic properties (Sachan and Karnwal, 2022). A low-fat/CN ratio permits fat retention inside the cheese and enhances its recovery from milk (Kalit et al., 2021). The lactose content directly alters the cheese's pH, enabling solubilization of higher amounts of calcium phosphate, increasing the moisture content of cheese, and thereby increasing the residual activities of chymosin and plasmin, which might lead to significantly altering the final texture and flavor (Ibáñez et al., 2021). Finally, CaCl_2 improves milk's rennet coagulation and syneresis during the cheesemaking process (McSweeney, 2007). The amount of these components is variable and highly dependent on the animal's stage of lactation, the plane of nutrition, the somatic cell count, and the season (Guinee et al., 2006). Therefore, some milk components must have a constant composition to produce cheese with the intended properties.

Fat is the most variable cheese-related constituent in milk, and its ratio to CN is the most critical composition variable to modify while standardizing milk for cheesemaking (Kalit et al., 2021). This standardization usually requires increasing the PFR, whose value depends on the type of cheese produced (Hill and Ferrer, 2021). For example, Mozzarella cheese requires a PFR of approximately 1.22, Parmesan of 2.02, and Cheddar of 0.91 (Hill and Ferrer, 2021). Adjusting the PFR ratio may involve the addition of skimmed milk, cream, or milk skimming (McSweeney, 2007; Hill and Ferrer, 2021). Additionally, the total protein content can also be adjusted through membrane filtration techniques. These techniques allow the retention of proteins and fat while permeating water, bacteria and their spores, soluble salts, and nonprotein nitrogenous compounds (Sachan and Karnwal, 2022). In consequence, the total protein content of milk can be increased from approximately 3.4% to 4.0%–5.0% (w/w) (Soodam and Guinee, 2018).

Different filtration techniques can be used for this purpose, including microfiltration, ultrafiltration, nanofiltration, and reverse osmosis (Sachan and Karnwal, 2022). These techniques are different in terms of the substances permeated, a factor that depends on the type of cheese being manufactured (McSweeney, 2007). Finally, the use of filtration techniques has been shown to increase the throughput of the process, because a higher amount of cheese can be produced

without increasing equipment sizes (Soodam and Guinee, 2018). In terms of cheese yield, higher retention of fat and proteins during the production process has been related to milk filtration techniques (McSweeney, 2007). A higher concentration of proteins allows for enclosing increasing amounts of fat during the clotting of cheese, and also whey can be retained in the cheese matrix, producing a greater mass of cheese per liter of milk only in some types of cheese (e.g., Parmesan and Swiss [Soodam and Guinee, 2018]) (Chamberland et al., 2022; Kalit et al., 2021).

12.3.2 Pasteurization

The pasteurization of milk is a procedure in which milk is heated to a specific temperature for a given period to reduce the pathogenic bacteria load (Franzoi et al., 2022). The heating procedure may involve keeping a temperature of 63°C for 30 min or 72°C for 16 s (Hill and Ferrer, 2021). This step is not mandatory in cheese fabrication. For example, cheeses such as the Colombian Costeño Cheese (Serpa Fajardo et al., 2016), Greek Kashkaval Cheese (Pappa et al., 2019), Grana Padano (Mammi et al., 2018), and Parmigiano Reggiano (Buonaiuto et al., 2021) are fabricated without pasteurization. In contrast, cheeses such as Cheddar and American soft artisan use this process (Franzoi et al., 2022). Although pasteurization provides safety to the milk and the produced cheese through pathogen elimination, cheese's structural and sensory properties may be detrimental. Heating milk to temperatures above 55°C causes the denaturation of whey to the extent that largely depends on temperature and time (Blecker et al., 2012). The denaturated whey sticks to κ -CN through sulfhydryl–disulfide interactions; therefore, enzymes face difficulties accessing κ -CN, resulting in a lengthy coagulation process (Franzoi et al., 2022). Additionally, the curd formed is weaker, and the cheese flavor develops slower and less intense because of the concomitant reduction of some microbial cultures contributing to these sensory properties (Fox et al., 2016).

12.3.3 Acidification

Acidification is a process in which the lactose contained in milk is converted into lactic acid through fermentation by lactic acid bacteria (LAB) (McSweeney, 2007). This process reduces the milk's pH from 6.6 to 4.2 (Sachan and Karnwal, 2022). This has consequences such as the precipitation of CN, allowing curd formation and water loss. Additionally, the acidification of milk avoids the growth of other microbe cultures and pathogens that are harmful to human health (Hill and

Ferrer, 2021; Vinicius De Melo Pereira et al., 2019). The rate and time at which the acidification process takes place primarily affect the activity of the coagulant, the microorganism's growth rate, coagulum strength, the syneresis process, the solubility of colloidal calcium phosphate, the cheese yield, and the rheological properties of the product (Bekele et al., 2019). For all these reasons, selecting the LAB starter culture is a critical step, and it is vital to ensure characteristics such as rapid acidification, dominance over other microbial cultures, production of good sensory properties in the final cheese, and survival in acidic conditions (Tidona et al., 2020). Different classifications of starter cultures may be used for the cheesemaking process, including liquid, frozen, and dried presentations (Farkye, 2004). LAB cultures can also be divided into mesophilic and thermophilic depending on the growth temperature. In the former, cultures thrive at temperatures between 10°C and 40°C (with an optimal range between 20°C and 32°C) (Serpa Fajardo et al., 2016), but in the latter, they bloom at temperatures between 20°C and 50°C (with an optimal range between 37°C and 45°C) (Johnson, 2013). Additionally, cultures of single strains or blends of cultures may provide different advantages to cheese production in terms of the uniformity of the acid development, flavor, and resistance to phages (Hill and Ferrer, 2021).

The original process to incorporate starter cultures in fresh milk involves “backslopping,” which consists of removing a small portion of the previous successful fermentation to inoculate the new one (Whittington et al., 2019). The type of starter culture is not the only factor that influences the pH value during this stage. The initial lactose concentration dictates the final amount of lactic acid produced by the fermentation and, therefore, the final pH value and cheese properties. These parameters vary as a function of the type of cheese manufactured. The excess lactose can be removed by washing the curd with water when the extent of the fermentation is achieved, while the lack of lactose can be addressed by adding lactose powder (Moynihan et al., 2016). Other microorganisms, also known as secondary cultures, may be added to cheese for ripening and provide some sensory properties by producing aromatic compounds such as diacetyl, acetoin, and butanediol (Sachan and Karnwal, 2022). High pH values during this stage produce soft, soapy, and bitter cheeses, while low ones produce brittle cheeses with a molten color (Hill and Ferrer, 2021). At low pH values, the electrostatic repulsion between CNs is reduced and, therefore, the rennet coagulation time. This reduction is exacerbated by the higher amount of Ca^{+2} solubilized in lower pH-value environments, allowing cheeses to achieve an increased firmness (McSweeney, 2007).

12.3.4 Coagulation

The enzymatic hydrolysis of κ -CN starts the coagulation process. κ -CN is mainly located on the surface of the CN micelles, as stated in Section 12.1. This location prevents CN micelle aggregation by electrostatic and steric effects under the physiological conditions of milk (i.e., $T < 37^\circ\text{C}$, $6.6 < \text{pH} < 6.7$ and $2 \text{ mM} < \text{Ca}^{+2} < 4 \text{ mM}$) (Guinee, 2021). The electrostatic repulsion of CN micelles is produced by the negatively charged C-terminal of the κ -CNs (fragment 106–169) present in the surface that protrudes from the surface of the micelles (Dagleish and Corredig, 2012). These protrusions are like hair threads perpendicular to the micelle surface that also provide steric hindrance when other micelles approach for aggregation, as shown in Fig. 12.2 (Glantz et al., 2010). CNs are hydrolyzed by several enzymes, such as calf chymosin, bovine and porcine pepsins, and fermentation-derived chymosins (Farkye, 2004). These enzymes, frequently referred to as rennet, may be of microbial, animal, or plant origin, and may vary in the optimal pH for hydrolysis and the cleavage site of CN (Farkye, 2004; Rajagopalan et al., 2021). For example, calf rennet typically hydrolyzes κ -CNs in the bond $\text{F}^{105}\text{-M}^{106}$ during coagulation, while the plant-derived enzyme *Wrightia tinctoria* serine protease in the bond $\text{N}^{123}\text{-T}^{124}$ (Rajagopalan et al., 2021). The hydrolysis of κ -CN in the bond $\text{F}^{105}\text{-M}^{106}$ produces two peptides, the hydrophobic (f1–105), also regarded as para- κ -CN, which remains in the curd, and the hydrophilic peptide f(106–169), which migrates to the whey phase (Farkye, 2004). The release of the peptide f(106–169) from the surface of the κ -CN relaxes the steric hindrance of the micelles (Britten and Giroux, 2022). Furthermore, the zeta potential of micelles reduces from 20 to 10 mV (indicating a reduction of the micelles' electrostatic repulsion) (McSweeney, 2007). Consequently, CN micelles start aggregating and forming a gel network or curd.

Although the proteolysis of κ -CN is crucial for clotting formation, the low degree of hydrolysis of the remaining CN fractions also aids in forming a firm curd (Bruno et al., 2010). Different elements, such as micelle size, pH, and composition variables, have been shown to alter the properties of curds (i.e., protein, fat, CN, and calcium content). Smaller micelles have been associated with firmer curds, but the protein content, fat, CN, calcium, pH, and CN-to-whey ratio influence curd properties and cheese processing (Glantz et al., 2010). Finally, milks with a high amount of protein tend to retain more fat and, in some cases, whey. For this reason, high yields, which usually correspond to 9%–15% (w/w), can be achieved with high protein milks (Farkye, 2004).

12.3.5 Whey drainage and salting

During this stage, whey is expelled through the cheese curd's shrinkage, a process called syneresis (Lucey, 2011). The curd is so stable over time after coagulation that the separation of the whey phase from the curd phase may be lengthy. For this reason, different processes are used to promote this expulsion, such as cutting, cooking, and pressing (Lucey, 2011; Fox et al., 2016). Curd cutting allows releasing the water and whey inside the gel matrix. For this reason, longer and faster cutting increase this separation (Hill and Ferrer, 2021). Curd cooking, which usually takes place after cutting, favors syneresis because higher temperatures cause bonds to break and, therefore, disrupt cavities within the curd, easing the water and whey expulsion and producing gels with high firmness (Fagan et al., 2007). Other factors affect the whey drainage in the curd. The cooking temperature influences the activity of the starter cultures depending on their type (i.e., mesophilic vs thermophilic) and, consequently, the acidification process, which alters the cheese's properties (Hill and Ferrer, 2021). High acidification in this stage, or lower pH values, directly correlates with higher syneresis because of CNs' proximity to the isoelectric point where they tend to aggregate (Fox et al., 2016). Additionally, curd cooking also inactivates the rennet and destroys most microbial flora, contributing to cheese ripening (Kaminarides et al., 2015).

Pressing cheese into molds is another process applied for whey drainage when most of the whey has already been expelled (McSweeney, 2007; Fox et al., 2016). Pressures up to 170 kPa are used in cheeses such as Cheddar, while for soft cheeses such as Blue and Argentinean Cremoso pressuring is not necessary (Hill and Ferrer, 2021; Cantor et al., 2017; Vélez et al., 2015). Finally, the increase of calcium concentration during syneresis has been found to increase whey separation at lower concentrations ($<10 \text{ mM}$), while at higher ones, such a trend is reversed (Marshall, 1982). The explanation for these observations is complex because of the interplay of different factors such as ionic strength, colloidal calcium phosphate solubilized, and pH (Fagan et al., 2007). Salting serves other purposes in cheese making, such as reducing the rennet activity during ripening, promoting whey drainage, CN hydration, protecting the cheese against starter and nonstarter bacteria, and, evidently, contributing to the salty flavor (Diaz-Bustamante et al., 2020; Guinee and Sutherland, 2016). The salt addition may involve immersing the molded curd into brines with NaCl concentrations of 18–25 (w/w) for $\frac{1}{2}$ to 5 days, adding salt directly to curds under constant mixing for 10–30 min before pressing and rubbing salt to the molded curds (Guinee and Sutherland, 2016).

12.4 Influence of casein polymorphisms on cheesemaking

CN polymorphisms correspond to different versions of the same CN fractions with variations in their amino acid sequences (Lisson et al., 2013). Because protein synthesis is addressed by the nucleotide sequence in the gene that encodes each protein fraction, polymorphic CNs or protein variants are produced in animals that include these gene variations. For this reason, most of the studies about the effects of CN polymorphisms on cheesemaking have involved the selection of a large number of animals, followed by the characterization of their milk to identify the polymorphisms of interest, and finally, the cheeses' production and their property evaluation (Bonfatti et al., 2011; di Gregorio et al., 2017; Ren et al., 2013). This experimental approach allows studying the effects of genetic variants directly on cheese production variables and functional properties. However, it makes it challenging to elucidate the sole impact of the molecular structure of the protein variant because gene variations not only produce proteins with different sequences but might also influence other compositional variables of milk. For example, gene variants may produce different milk protein compositions (Bonfatti et al., 2010; Poulsen et al., 2017a), fat contents, and Ca/protein ratios (Ren et al., 2013).

Consequently, the cheese properties changes can be attributed to compositional changes and not necessarily to the production of a protein variant with a different molecular structure. Conversely, thus far, most studies have mainly focused on building relationships between the presence of a particular genetic polymorphism in some breeds and its effects on milk composition and, ultimately, on cheesemaking traits. To illustrate this, the relationships between gene CN polymorphisms and cheesemaking will be addressed for cattle. Importantly, the same approach has been used for other mammals such as sheep (Corral et al., 2013; Sinaga et al., 2016), camel (Shuiep et al., 2013), and buffalo (Zicarelli et al., 2020).

Most studies on polymorphic variants of CNs have focused on fractions β -CN and κ -CN since they are the most influential on milk's nutritional traits and the properties of the derived products (Lisson et al., 2013).

12.4.1 Influence of β -casein on cheesemaking

The influence of β -CN in the coagulation of milk remains elusive. In terms of the variants β -CN A¹ and β -CN A², the results yield different conclusions. Some studies report the production of higher amounts of β -CN with the allele β -CN A² (Heck et al., 2009), while

others describe the opposite (Nadugala et al., 2022; Morris et al., 2010). This difference may be due to the influence of other simultaneous polymorphisms in a different protein or the effect of other parameters such as the animal's season, age, and breed (Bonfatti et al., 2010). When allele β -CN A¹ was related to the decrease of α 2-CN and β -CN in comparison to β -CN A², an increase in the ratio of α 1-CN and κ -CN fractions was also observed (Bonfatti et al., 2010). This increment in the κ -CN content with allele β -CN A¹ may explain the improvement of milk coagulation properties compared to β -CN A² (Daniloski et al., 2022a). In this regard, the milk coagulation mechanism indicates that large amounts of κ -CN are directly related to smaller CN sizes, higher capacity of aggregation of micelles, and the formation of denser and less porous gels with allele β -CN A¹ milks (Daniloski et al., 2022a; Abeykoon et al., 2016). Additionally, better coagulating milks are usually rich in ordered secondary structures such as β -turn and α -helix, which are absent in β -CN A² (i.e., random structures) (Daniloski et al., 2022a, 2022b). This is an example of how milk's coagulation to produce cheeses relates to the κ -CN content, polymorphisms, and the β -CN variants.

The genotype β -CN A²A² produces higher fat yields than genotypes β -CN A¹A¹ and β -CN A¹A² (Ristanic et al., 2021), while there was a higher coagulation time and lower curd firmness of those with the allele A² in comparison to alleles β -CN A¹A¹ and BA¹ (Hallén et al., 2007; Bisutti et al., 2022). These results point to the importance of the allele β -CN A¹ in cows that produce milk appealing for cheesemaking. Although the type of milk suitable for making a specific variety of cheese depends on the target properties of that cheese, good coagulation properties are directly related to high yields because firmer gels can retain more elevated amounts of other substances, thus favoring the economic feasibility for cheesemaking (Abeykoon et al., 2016). Concerning allele β -CN B, it has also been shown to enhance coagulation properties (i.e., lower coagulation times, higher curd firmness, and higher coagulum yield) in different studies in comparison to allele β -CN A² because of the presence of a higher amount of κ -CN (Abeykoon et al., 2016; Lucey, 2011). Regarding milk protein composition, β -CN I has been related to a higher percentage of β -CN and a lower percentage of α 1-CN compared to alleles A¹ and A² (Bonfatti et al., 2010). Concerning milk coagulation, variant β -CN I was only associated with variant κ -CN B in the composite haplotype β - κ -CN IB and it provided higher protein yield (Visker et al., 2011), as well as higher protein percentage and curd firmness (Vallas et al., 2012). This result is not conclusive for variant β -CN I because these effects can also be related to the presence of κ -CN B and is not exclusively inherent to

variant β -CN I. This is supported by the similar magnitudes of protein yield that can also be obtained in combinations of variant β -CN A² with κ -CN B in the haplotype β - κ -CN A²B (Visker et al., 2011). Variant β -CN I has been indistinctly found in milk with good and poor coagulation properties (Jensen et al., 2012a). Variant β -CN F has been only correlated with noncoagulating milks (Poulsen et al., 2017a). Although the relationship between allele β -CN F and the amount of β -CN has not been evident, a relationship with the amount of κ -CN may exist due to the suggested presence of small proportions of κ -CN in noncoagulating milks (Poulsen et al., 2017a).

12.4.2 Influence of κ -casein on cheesemaking

The fraction κ -CN is related to the CN sizes and their tendency for aggregation (Abeykoon et al., 2016). Because CN micelles significantly influence the coagulation mechanism process, the κ -CN polymorphisms have been shown to primarily affect cheesemaking, especially in the rennet coagulation (Nadugala et al., 2022). Since 1967, the phenotypes containing the allele κ -CN B, such as κ -CN AB and κ -CN BB, have been related to shorter rennet clotting times, curd firming time, higher curd tensions, curd, and milk yields, and curd firmness compared with phenotypes κ -CN AA, AB, AE, and BE (Hallén et al., 2007; Sherbon et al., 1967; Jensen et al., 2012a; di Gregorio et al., 2017; Zicarelli et al., 2021). Reductions of up to 24% of clotting time and increments of up to 37% of the maximum achievable curd firmness were observed for milk with phenotype κ -CN BB in comparison to κ -CN AA, a fact that was attributed to the higher concentrations of κ -CN and α 2-CN and to a lesser extent, the presence of α 1-CN, for the milk BB (Rahali and Ménard, 1991). Therefore, thus far, it is unclear if the improved cheesemaking properties of κ -CN B result from higher κ -CN concentration, variation in the ratios of other CN fractions, or changes in micelle properties (Heck et al., 2009). However, when the concentration of proteins was kept constant, the allele κ -CN B also exhibited the same rennet clotting time and curd tensions (Jakob and Puhan, 1992). The improvement of coagulation properties seems not to be related to the total amount of CN but to the relative proportion of the fraction κ -CN (Poulsen et al., 2017b). Allele κ -CN B correlates well with a higher κ -CN percentage at the expense of the detriment of other fractions, such as α 1-CN, which suggests that proteins compete for expression in the mammalian gland (Bonfatti et al., 2010).

Concerning the total amount of protein, some studies have related it to shorter coagulation time and higher firmness. In contrast, others have reported

different tendencies, and today this relationship remains elusive, thus supporting the idea that coagulation properties seem to depend on the quality of the protein and not so much on the quantity (Hallén et al., 2007). In terms of heat stability, milk with allele κ -CN AA pasteurized at 72°C for 15 s showed a 47% increase in cutting time compared to milks with phenotypes that include allele B, thus indicating that referring to cheesemaking, heat has greater adverse effects on milk with allele A (i.e., longer process) (Jakob and Puhan, 1992). The impact of the polymorphism κ -CN E on the renneting properties was observed to be similar to the polymorphism κ -CN A, and both are related to poor coagulation properties (i.e., longer rennet coagulation time, lower maximum coagulum strength by storage modulus G', and lower curd firming rate) (Jakob and Puhan, 1992; Jensen et al., 2012a). However, between alleles κ -CN A, κ -CN B, and κ -CN E, a study shows allele κ -CN E as producing the worst coagulation properties with a longer rennet coagulation time and lower curd firming rate (Poulsen et al., 2017b). Finally, the higher hardness, stretchability, and chewiness for haplotype AB cheeses compared to AE and EB may be due to compositional aspects, such as the CA/protein ratio, fat, and moisture (Ren et al., 2013).

In terms of α 1-CN, most of the studies analyzing the influence of its polymorphisms on cheese properties have been conducted through composite genotypes, as shown in Section 12.3.3. However, variant α 1-CN A, which has a deletion of 13 amino acids, has led to softer curds than variant B, whose deletion is not present (Farrell et al., 2004a). The deleted moiety is highly hydrophobic, therefore, causes a weaker interaction of this CN fraction during chymosin cheese ripening (Farrell et al., 2004a).

12.4.3 Composite genotypes

Numerous compositional variables (e.g., salts, fat, proteins) have made it challenging to examine the effect of a single CN polymorphism on cheesemaking. Additionally, other gene polymorphisms, some probably related to proteins other than CNs, might be expressed simultaneously. For instance, a study of the effect of allele β -CN A¹ on cheese properties can provide wrong insights if failing to monitor the presence of polymorphisms in other CN fractions (i.e., α 1-CN, α 2-CN, κ -CN). Thus, the correlations between β -CN A¹ and cheese properties could overlap with the simultaneous presence of other gene polymorphisms, especially if cattle sampling is limited. To partially address this difficulty, some researchers have decided to study not the presence of one polymorphism but the simultaneous occurrence of different CN alleles.

Polymorphisms of two CN fractions κ -CN (A, B, and E) and β -CN (A^1 , A^2 , and B) were studied simultaneously, suggesting the combination β - κ -CN A^2B as the most suitable in terms of high protein yield and cheese production (Heck et al., 2009). The analysis of polymorphisms by combining three CN fractions has been developed to study the properties of milk and cheese. For example, the haplotype $\alpha 1$ - β - κ -CN with the alleles B- A^1 -B has been linked to higher milk yields and fat and protein percentages (Nadugala et al., 2022). Consequently, milk with higher protein favors higher yields in cheesemaking. In the study of the complete diploid combinations of $\alpha 1$ - β - κ -CN, the composite genotype BB- A^1A^2 -AB showed to produce milk of Swedish Red and Danish Holstein cattle with higher relative concentrations of β -CN and κ -CN and lower α -CN (Gustavsson et al., 2014). This increase, especially of κ -CN, could be related to better coagulation properties for cheesemaking in comparison to the other studied combinations of fractions alleles of β -CN (A^1 , A^2 , B, and I) and κ -CN (A, B, and E) (Gustavsson et al., 2014). In this case, the genotype BB- A^2A^2 -AA showed the opposite results for the relative concentration of CN fractions, probably due to the absence of the alleles β -CN A^1 and κ -CN B (Gustavsson et al., 2014). The genotype BB- A^2A^2 -AA was also confirmed as producing poor coagulating or noncoagulating milks in Danish Jersey and Holstein Friesian cows (Jensen et al., 2012b). The noncoagulating milks were observed to have a direct relationship with a lower phosphorylated fraction of κ -CN (Jensen et al., 2012b). As discussed above, changes in the amino acid sequences of polymorphisms influence the number of phosphorylation sites in the protein. Thus, it is expected that κ -CN polymorphisms with fewer phosphorylation sites produce undesirable cheesemaking properties, such as poor coagulation capacity. In another study, the genotype $\alpha 1$ -CN BB was maintained constant while exploring variations of β -CN (A^1 and A^2) and κ -CN (A and B) for Italian Holstein cows (Perna et al., 2016). As a result, the genotype BB- A^1A^1 -AA showed the highest fat content, while the genotype BB- A^2A^2 -BB showed the highest protein content, therefore, better coagulation properties and higher yields (Perna et al., 2016). These results correlate well with the necessity to have allele κ -CN B to ensure a highly coagulating milk (Gustavsson et al., 2014); however, the need to simultaneously have allele β -CN A^1 has not been confirmed because the study failed to consider combinations of allele κ -CN B with allele β -CN A^1 (Perna et al., 2016). Coagulation traits are apparently dependent on the type of coagulation (i.e., rennet or acid). Thus, the composite $\alpha 1$ - β - κ -CN genotypes BB- A^1A^2 -AA, BB- A^1A^2 -BE, BB- A^2A^2 -AA, BB- A^2A^2 -BB, BC- A^2A^2 -BB were tested, demonstrating the lowest rennet

time formation for genotype BC- A^2A^2 -BB and the lowest acid gelation time for genotype BB- A^2A^2 -AA (Ketto et al., 2017). However, in both coagulation tests, the genotypes BB- A^2A^2 -BB and BB- A^1A^2 -BE showed the longest coagulation times (Ketto et al., 2017), a result inconsistent with another study that reported on the good coagulation properties of the composite genotype BB- A^2A^2 -BB (Perna et al., 2016). Milk polymorphisms are full of inconsistencies regarding the experimental evidence. For example, one study says CN haplotypes fail to impact milk traits (i.e., fat, protein yield, and fat/protein content) (Bonfatti et al., 2010). In contrast, another study says that CSN1S1 C has a considerable influence on milk protein content (Boettcher et al., 2004). One study reports higher fat yield for β -CN A^2A^2 over other polymorphisms evaluated (Ristanic et al., 2021), while another study found no significant influence of β -CN genetic variants on milk traits (Braunschweig, 2008). This can be explained by the possible association of breeds and sire families (Bonfatti et al., 2010), which are different between both studies, that is, Serbian Holstein Friesian and Swiss Fleckvieh cattle (Ristanic et al., 2021; Braunschweig, 2008).

12.5 Conclusions

CN polymorphisms are different versions of the same base protein fraction (i.e., $\alpha 1$ -CN B, $\alpha 2$ -CN A, β -CN A^2 , and κ -CN A) with some sequence changes that involve substitution, deletion, and insertion of amino acids (Martin et al., 2013). They are important due to strong correlations with milk traits and derived products (Soloshenko et al., 2016). The polymorphisms of a protein are named in alphabetic order depending on the year they were discovered. Eventually, subscripts of the same letter in numerical ranking can be used to differentiate specific fractions because of the simultaneous assignation of the same letter by two authors and based on their mobility in assays involving solid-phase and porous systems through which molecules migrate (Farrell et al., 2004a). CN polymorphisms influence the cheese production process in different stages ranging from raw milk to CN crosslinking and whey drainage to obtain the final cheese. During these stages, CN polymorphisms influence various coagulation processes (i.e., acid and rennet coagulation) by modifying parameters mainly related to time (Ketto et al., 2017), CN micelle size, compositional traits (i.e., fat content, total protein content, the composition of CN fractions, and Ca/protein ratio), and CN crosslinking to form the curd. As a consequence, curd rheology (i.e., G' and G'' moduli) (Jakob and Puhani, 1992; Jensen et al., 2012a), the overall yield

of the process, and the hardness and firmness of cheese are also altered (McSweeney, 2007; Ren et al., 2013; Perna et al., 2016).

Studies conducted on CN polymorphisms have focused on identifying the presence of a protein variant in milk and evaluating cheeses produced from it. The presence of a specific polymorphic variant means that the animal of origin has the polymorphic gene that encodes that protein. In this sense, the sole influence of a variant protein with a different molecular structure on cheese properties has been impeded because gene variations do not only allow producing a different protein but also affect other milk compositional variables such as the fat content, protein content, CN fractions ratios, and Ca/protein ratio (Ren et al., 2013; Bonfatti et al., 2010; Poulsen et al., 2017a). Consequently, studies have been limited to finding relationships between the gene polymorphism and milk and cheese properties. Even though gene polymorphisms strongly influence milk properties, these relationships are still superficial and ignore other potentially influential factors, such as the presence of protein polymorphisms in different CN fractions and other proteins, such as lactoglobulin. This influence was addressed by studying multiple protein polymorphisms at the same time (i.e., composite polymorphisms), so the presence of polymorphisms in two and three CN fractions was considered simultaneously (Heck et al., 2009; Ketto et al., 2017). However, a thorough mechanistic understanding of the influence of CN polymorphisms derived from gene polymorphisms is necessary to obtain a deeper insight into their effects on cheesemaking. One way of doing this would be to attempt to elucidate the behavior of purified CN polymorphisms in terms of their specific interactions with each type of milk component. Then, relate these behaviors at a macroscopic and microscopic scale (e.g., particle diameter, viscosity, firmness, taste) to the possible underlying phenomena at a molecular scale via zeta potential, mass spectrometry, pH, near infrared spectra and nuclear magnetic resonance. From this, it is clear that milk is a high complexity system, therefore studying as a whole could result in ambiguous preliminary results. This knowledge gap has resulted in contradictory findings in some studies, because milk properties are affected not only by polymorphisms but also by the animal's age, nutrition, season, and breed (Guinee et al., 2006; Bonfatti et al., 2010).

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CASEIN

Structural Properties, Uses, Health Benefits and Nutraceutical Applications

Edited by
Mamdouh El-Bakry
Bhavbhuti M. Mehta

An exploration of casein, its types, properties, and applications in the food and pharmaceutical industries

Casein: Structural Properties, Uses, Health Benefits and Nutraceutical Applications investigates casein properties, uses, and applications in food and non-food products, in addition to exploring its health benefits and uses in manufacturing, such as in cheese products, along with an in-depth discussion on the future scope, challenges, and market trends of this protein.

Casein: Structural Properties, Uses, Health Benefits and Nutraceutical Applications is an excellent reference for food scientists, dairy researchers, pharmaceutical scientists, students and researchers studying related fields.

Key Features

- Provides comprehensive coverage of casein, the main milk protein that has many applications and uses
- Includes suggested reading for further information
- Addresses a wide-range of related topics, including non-food applications of casein

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